PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

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Email: umhackathon@um.edu.my

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Project Name: Podpilot

Version: 1.0

1. Project Overview

- **Problem Statement:** Malaysian print-on-demand (POD) sellers on platforms like Shopee and Etsy struggle with fragmented, manual workflows across the entire product lifecycle. They must independently track trends (e.g., from TikTok), design products, create listings, respond to customer issues, and manage finances using disconnected tools. This leads to inefficiencies, slow time-to-market, inconsistent decision-making, and limited scalability for small sellers.
- **Target Domain:** AI Systems & Agentic Workflow Automation, specifically applied to e-commerce enablement for small-to-medium POD businesses in Malaysia.
- **Proposed Solution Summary:** PODPilot is a GLM 5.1-powered multi-agent AI system that acts as an autonomous business co-pilot for POD sellers. It ingests unstructured inputs and orchestrates specialized agents to handle the full lifecycle: from business ideation and product design to automated publishing via Printify, customer service response, and financial tracking. By maintaining a shared session state across agents, PODPilot enables seamless, end-to-end automation with intelligent decision-making (e.g., pricing, margin optimization), significantly reducing manual effort while improving operational efficiency and scalability.

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2. Background & Business Objective

- **Background of the Problem:**

  The rise of print-on-demand (POD) platforms like Printify has lowered the barrier to entry for e-commerce, enabling individuals to start online businesses without inventory. At the same time, marketplaces such as Shopee and Etsy, along with trend sources like TikTok, have accelerated product discovery and demand cycles.

  However, while access has improved, operations remain highly fragmented. Sellers must manually move between platforms for trend research, design creation, listing optimization, customer service, and financial tracking. These tasks require different skill sets and tools, leading to inefficiencies, inconsistent quality, and slow response to market trends, especially for small sellers with limited resources.

- **Importance of Solving This Issue:**

  Addressing this fragmentation is critical because speed and adaptability are key success factors in POD businesses. Trends can emerge and fade within days, and delays in product creation or listing can result in missed revenue opportunities. Additionally, poor handling of customer service or pricing decisions can directly impact seller ratings and profitability.

  Without automation and intelligent coordination, small sellers face scalability limits and are often outcompeted by larger, more resource-equipped sellers. Solving this issue enables sellers to operate more efficiently, make data-driven decisions, and remain competitive in a fast-moving digital marketplace.

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- **Strategic Fit / Impact:**

  PODPilot aligns with the growing adoption of AI-driven automation in e-commerce by introducing a fully integrated, multi-agent system that manages the entire product lifecycle. It transforms fragmented workflows into a unified, stateful process, enabling:

  - Faster time-to-market for trending products
  - Improved operational efficiency through automation
  - Smarter decision-making using AI-driven insights (pricing, margins, customer handling)
  - Scalable business operations without proportional increases in workload

  The impact is particularly strong for the Malaysian market, where many small and solo entrepreneurs rely on platforms like Shopee. By reducing operational complexity and enhancing productivity, PODPilot empowers sellers to grow sustainably while positioning itself as a scalable AI solution for the broader e-commerce ecosystem.

3. Product Purpose

3.1. Main Goal of the System: The primary goal of PODPilot is to function as an autonomous, end-to-end business co-pilot that streamlines and automates the entire print-on-demand (POD) workflow. It leverages multi-agent AI coordination to transform unstructured inputs into

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actionable outputs—covering business ideation, product design, listing creation, customer service handling, and financial tracking.

By integrating directly with platforms like Printify, Etsy, and Shopee, the system minimizes manual intervention, reduces operational friction, and enables faster, data-driven decision-making. Ultimately, PODPilot aims to improve seller productivity, shorten time-to-market, and maximize profitability through intelligent automation.

3.2. Intended Users (Target Audience):

- **Solo entrepreneurs and side hustlers** who lack time or expertise to manage multiple aspects of a POD business
- **Small business owners** seeking to scale operations without increasing manpower
- **Beginner sellers** who need guidance in branding, product selection, and pricing strategies
- **Content-driven sellers** leveraging platforms like TikTok for trend discovery but lacking structured tools to act on those trends

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4. System Functionalities

4.1. Description: PODPilot operates as a multi-agent orchestration system powered by GLM 5.1. An Orchestrator acts as the central brain — it classifies any unstructured input (TikTok URL, voice note, screenshot, or vague text prompt) and routes it to the appropriate specialized agent. Full session state is maintained across all agents using Redis, enabling seamless handoffs through the entire product lifecycle.

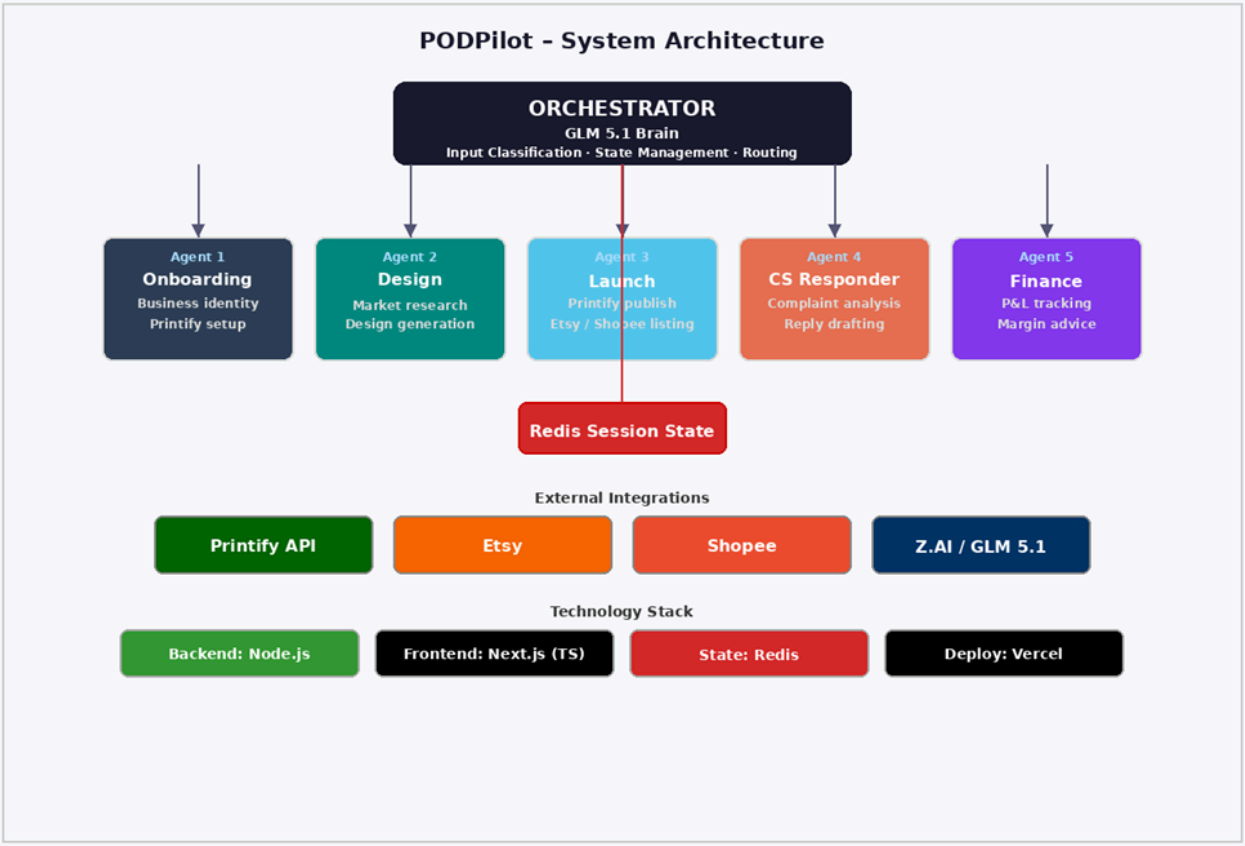

Figure 4.1: PODPilot System Architecture

4.2. Key Functionalities:

Orchestrator — GLM 5.1 Brain

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- Accepts any unstructured input: TikTok URL, voice note, screenshot, plain text
- Classifies intent and routes to the correct agent
- Maintains full session state across the entire lifecycle via Redis

Agent 1 — Onboarding Agent

- Receives a business idea as the initial prompt
- Provides ideas to determine a business identity and direction
- Creates a Printify account programmatically aligned with the seller's brand

Agent 2 — Design Agent

- Receives a product prompt and performs market research to determine optimal pricing
- Generates or suggests viable product designs
- Recommends suitable product types based on demand and margin data

Agent 3 — Launch Agent

- Receives product information and suggested product types from the Design Agent
- Creates and publishes products on Printify via REST API
- Syncs listings to the connected Etsy or Shopee store automatically

Agent 4 — CS Responder

- Reads customer complaint screenshots or pasted messages
- Drafts tone-appropriate replies tailored to complaint severity

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- Flags refund or reprint actions with severity levels for seller review

Agent 5 — Finance Tracker

- Auto-logs P&L from every published product and every CS case resolved
- Uses GLM to generate actionable margin and marketing recommendations

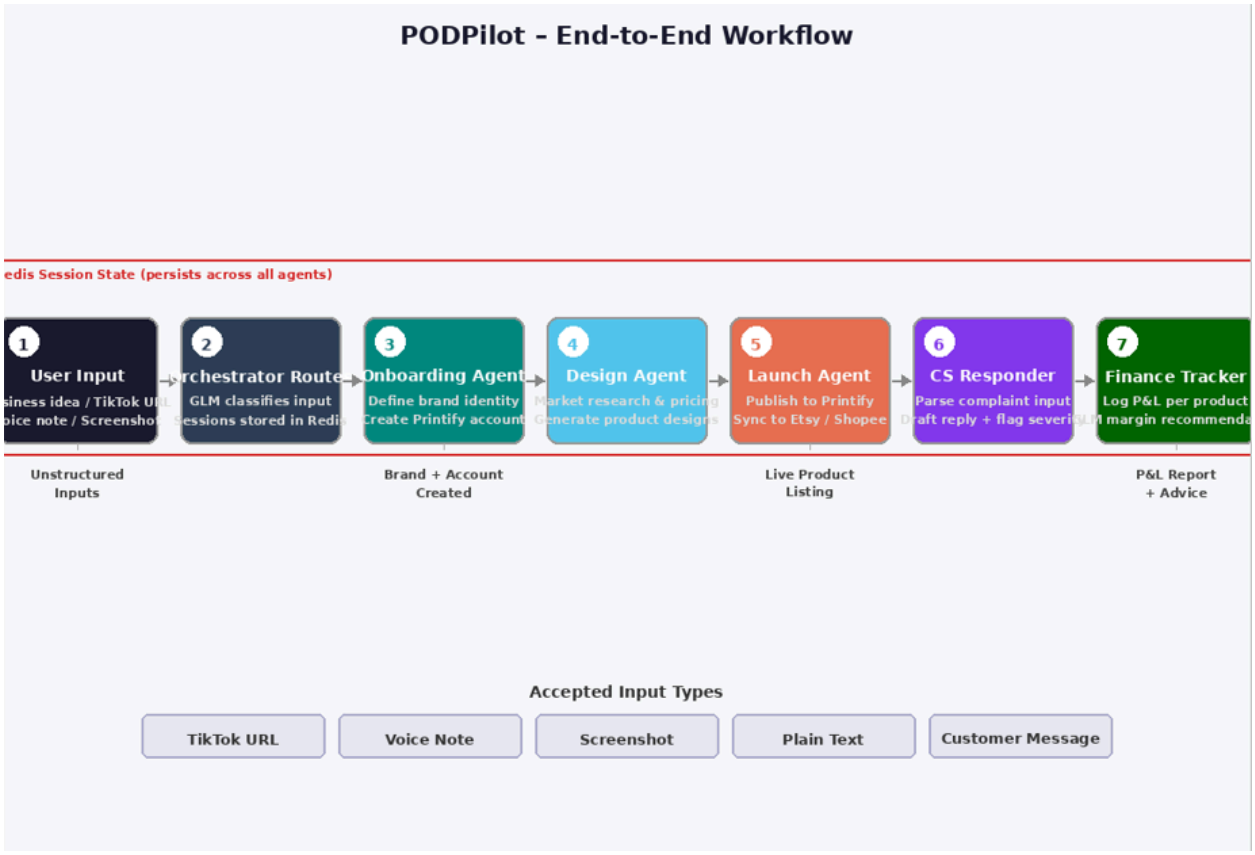

Figure 4.2: End-to-End Agent Workflow

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4.3. AI Model & Prompt Design

4.3.1. Model Selection:

PODPilot uses GLM 5.1 via the Z.AI API as its core LLM. GLM 5.1 was selected for the following reasons:

- Native support for multimodal inputs — handles text, images, and structured data, enabling the system to process TikTok screenshots, complaint images, and design prompts within the same model
- Strong instruction-following and JSON-mode capabilities, critical for structured agentic outputs (product specs, pricing JSON, severity flags)
- Competitive inference cost vs. capability trade-off appropriate for the Z.AI API credits available during the hackathon
- Contextual reasoning strength suited to pricing rationale and margin advice — genuine decision-making, not just text generation

4.3.2. **Prompting Strategy:**

PODPilot uses a multi-step agentic prompting strategy combining:

- System-level role assignment per agent (e.g., 'You are a POD market research specialist...')
- Few-shot examples embedded in the Orchestrator prompt to improve classification accuracy on edge-case inputs
- Chain-of-thought steps within the Design and Finance agents to produce justified, auditable outputs (e.g., pricing rationale, margin analysis)

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This strategy was chosen because it balances controllability with flexibility: each agent operates with a well-defined role and output schema, reducing hallucination risk while the few-shot context improves robustness on ambiguous real-world inputs.

4.3.3. **Context & Input Handling**:

- Maximum supported input per call: ~8,000 tokens for text; images are resized to ≤1024px before embedding
- Inputs exceeding token limits are chunked at sentence boundaries before being sent to the LLM
- Oversized images are auto-compressed to JPEG at 80% quality before processing
- Malformed or unrecognized inputs are returned to the Orchestrator with a fallback classification for human review

4.3.4. **Fallback & Failure Behavior:**

- Off-topic or hallucinated model responses are detected via output schema validation (JSON schema checks on structured agent outputs)
- On schema failure, the agent retries the prompt up to 2 times with a corrective suffix ('Return ONLY valid JSON matching the schema')
- If 3 consecutive failures occur, the session is flagged and the seller is prompted with a graceful error state and an option to rephrase their input
- Critical failures (e.g., Printify publish errors) trigger an alert in the Finance Tracker and are logged for review

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5. User Stories & Use Cases

- **User Stories:**

• As a solo POD seller, I want to describe a product idea in plain language so that PODPilot creates and publishes a listing on Etsy without me touching any design or publishing tool.

• As a small business owner, I want to paste a customer complaint screenshot so that PODPilot drafts a professional reply and tells me whether to issue a refund or reprint.

• As a beginner seller, I want to input a TikTok URL so that the system identifies a trending product opportunity and walks me through creating it.

• As a seller, I want to see a live P&L dashboard updated after each sale or CS case so that I can understand my margins and act on AI-generated recommendations immediately.

- **Use Cases (Main Interactions):**

Business Onboarding Flow

The seller enters a business idea prompt (e.g., 'I want to sell motivational gym apparel targeting Malaysian students'). The Onboarding Agent uses GLM to suggest a brand name, niche focus, and visual identity direction. A Printify account is created via API. The Design Agent is then triggered with an initial product suggestion from the Orchestrator.

Product Design & Launch Flow

The Design Agent performs market research to determine pricing benchmarks and demand signals. It generates design concepts and recommends product types (e.g., t-shirts, hoodies,

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mugs). Approved designs are passed to the Launch Agent, which calls the Printify REST API to create the product and sync it to the connected Etsy or Shopee store automatically.

Customer Service Flow

The seller uploads a complaint screenshot or pastes a message. The CS Responder parses the message using GLM, determines the issue type and severity level (Low / Medium / High), and drafts a tone-appropriate reply. Refund or reprint flags are surfaced for the seller's one-click approval.

Finance & Recommendation Flow

After each product publish or CS resolution, the Finance Tracker auto-logs the event to the P&L ledger stored in Redis. GLM analyses cumulative margin data and generates actionable recommendations (e.g., 'Increase base price of Product X by 15% to reach 40% margin at current fulfilment cost').

- **Use Case Diagram:**

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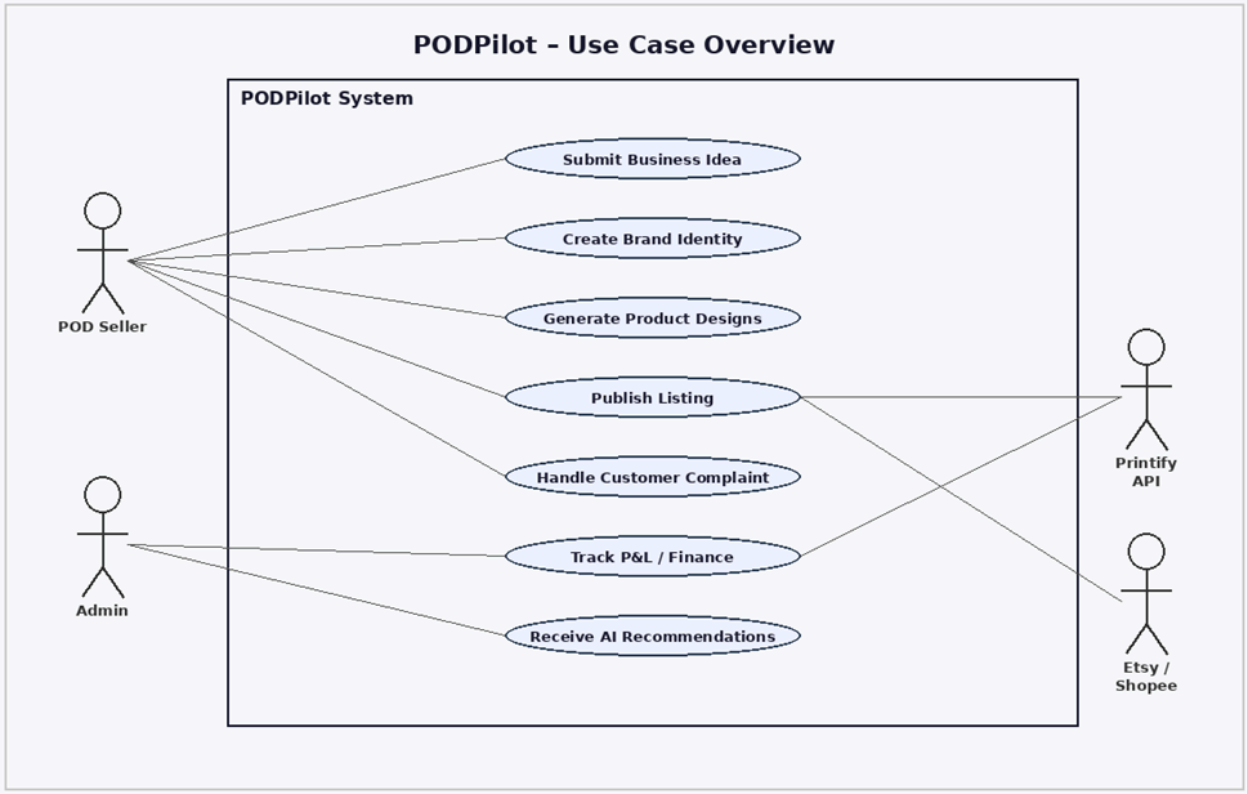

6. Features Included (Scope Definition)

- Multi-agent orchestration powered by GLM 5.1 via Z.AI API
- Business onboarding: brand ideation and automated Printify account creation
- Design Agent: market research, pricing recommendation, and AI-generated product design suggestions
- Automated product publishing to Printify with direct sync to Etsy (primary) and Shopee (secondary)
- Customer complaint intake via screenshot or pasted text, with AI-drafted replies and severity flagging

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- Real-time P&L tracking with GLM-generated margin and marketing recommendations
- Stateful session management across all agents using Redis
- Next.js (TypeScript) frontend dashboard with Finance panel and CS action center
- Node.js backend with REST endpoints for all agent interactions
- Deployment on Vercel

7. Features Not Included (Scope Control)

- TikTok Shop × Printify direct integration — US-only; Malaysian sellers use Shopee or Etsy instead (out of scope)
- Native mobile app — web-responsive only in this version
- Multi-user / team collaboration within a single PODPilot account
- Custom fulfilment partner integration beyond Printify (e.g., Printful, Gelato)
- Automated ad campaign creation or social media posting
- Real-time TikTok trend ingestion via TikTok API (requires TikTok developer approval; seller manually provides TikTok URLs)
- Production-grade authentication and RBAC beyond basic session handling
- Shopee OAuth integration in the initial demo — Etsy is the primary connected channel

8. Assumptions & Constraints

- **LLM Cost Constraint:**

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Estimated token cost per average user session: ~12,000 input tokens + ~3,000 output tokens across all agent calls, totalling approximately RM 0.08–0.15 per session at current Z.AI pricing. To keep inference costs manageable at scale, the system implements:

• Redis-based response caching for repeated market research queries (e.g., same product category queried within 24 hours reuses the cached GLM output)

• Prompt compression: conversation history is summarised rather than passed in full after 3 agent turns

- **Technical Constraints:**

  • Printify publish API is rate-limited to 200 requests per 30 minutes — bulk publishing is queued and throttled accordingly

  • Store connection to Printify (Etsy/Shopee OAuth) is a one-time manual onboarding step; all subsequent product publishing is fully automated via API

  • Design generation relies on GLM's image understanding and text-based design description; actual design file rendering is handled by Printify's own design layer

- **Performance Constraints:**

  • Target end-to-end latency from input submission to published listing: under 90 seconds for the demo flow

  • Maximum concurrent sessions supported in demo environment: 10 (Vercel hobby tier limits apply)

- **User Input:**

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- The system requires human approval before publishing any listing to Etsy or Shopee — no fully unsupervised auto-publish in v1.0
- Voice note inputs are transcribed client-side via browser Web Speech API before being sent to the Orchestrator
- Screenshot inputs must be under 5 MB and in JPEG or PNG format

9. Risks & Questions Throughout Development

Design Uniqueness

Risk: GLM may generate similar product design descriptions across different users querying the same trend niche. Mitigation: Each Design Agent call includes a uniqueness seed derived from the seller's brand identity and a random creativity temperature (0.8–1.0) to diversify outputs. Post-generation, a simple similarity check against recently generated designs in Redis cache flags near-duplicates for the seller to review.

Listing Compliance

Risk: Auto-generated Etsy/Shopee listings may violate platform content policies (e.g., prohibited keywords, trademark references). Mitigation: Listings pass through a lightweight GLM content moderation step before the Launch Agent submits them. Flagged listings are held in a 'review' state and surfaced to the seller.

Hallucinated Pricing Data

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Risk: GLM may generate plausible-sounding but inaccurate market pricing data. Mitigation: The Design Agent's market research step is grounded by fetching real Etsy/Shopee listing price ranges via a lightweight scraper. GLM then reasons over real data rather than generating pricing from memory. If the scraper fails, the agent discloses to the seller that pricing is AI-estimated and recommends manual verification.

CS Reply Tone Mismatch

Risk: GLM-drafted customer service replies may be too formal or too casual for a specific seller's brand voice. Mitigation: The CS Responder reads the seller's stored tone preference (set during onboarding) from Redis and includes it in the system prompt. Sellers can also edit drafts before sending.

Printify API Downtime

Risk: Printify API unavailability would block the Launch Agent. Mitigation: Failed publish attempts are queued in Redis and retried with exponential backoff (max 3 retries over 10 minutes). Sellers are notified of delays via the dashboard status panel.

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